# Bias Collapse in Shallow ReLU Networks

Caleb Abrantes, Bobby Chen, Om Kannan
NSF RTG

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# Motivation

### The overparameterization puzzle:

- Modern networks have far more parameters than the problem demands
- Classical learning theory predicts overfitting; practice shows robust generalization
- Why does the optimizer self-simplify?

### Bias collapse: a concrete mechanism

- Gradient flow automatically eliminates redundant neurons, with no regularization required
- $m = 1500$ neurons $\longrightarrow$ effective complexity of $k = 7$ neurons
- The number of surviving *bias clusters* is approximately around the number of *inflection points* of the target function

**This work**

Rigorous numerical investigation of three open problems:

1. **OP 4.1:** Does $C(m, f^*) \to k$ as $m \to \infty$?
2. **OP 4.2:** Does collapse generalize to $d \geq 2$ dimensions?
3. **OP 4.3:** Is there a provable $L^2$ pruning bound after collapse?

**72 runs,** 9 targets, $m$ up to 5000.

# Model and Gradient Flow

### 1D shallow ReLU network:

$$f(x) = \sum_{j=1}^{m} a_j \, \sigma(x - b_j), \quad \sigma = \max(0, \cdot), \quad x \in [-1, 1]$$

Each neuron places a *kink* at bias $b_j$ with slope change $a_j$.

**Gradient flow on MSE** $\mathcal{L} = \frac{1}{2} \int_{-1}^{1} (f - f^*)^2 \, dx$**:**

$$\dot{a}_j = -\int_{-1}^{1} (f - f^*) \, \sigma(x - b_j) \, dx$$

$$\dot{b}_j = a_j \int_{b_j}^{1} (f - f^*) \, dx$$

Solved via `scipy` RK45, $T = 500$, rtol $= 10^{-4}$.
Stationarity criterion: $\max_j |\dot{a}_j|, |\dot{b}_j| < 0.01$.

### Key measured quantity:

$$C(m, f^*) \; = \; \#\{\text{bias clusters}\}$$

at stationarity; gap threshold 0.02.

**Fixed-point condition**

Each *active* neuron ($a_j \neq 0$) satisfies $R_j := \int_{b_j}^{1} (f - f^*) \, dx \approx 0$ at stationarity. Verified numerically at every converged run; connects to the theory of OP 4.1.

# The Three Open Problems

### OP 4.1:Cluster Count

$$\lim_{m \to \infty} C(m, f^*) = k$$

Does the number of bias clusters converge to the number of sign-changing inflection points of $f^*$?

### OP 4.2:Higher Dimensions

For $f^* : [-1, 1]^2 \to \mathbb{R}$, does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

### OP 4.3:Pruning Bound

After collapse, is there a provable $L^2$ error bound when pruning to $k$ neurons?

$$\left\| \tilde{f} - f \right\|_{L^2} \leq \delta \cdot \sum_j |a_j|$$

Each open problem is introduced in detail immediately before its results.

# Open Problem 4.1: Cluster Count Conjecture

## Conjecture

$$\lim_{m \to \infty} C(m, f^*) \; = \; k$$

where $k :=$ # sign-changing inflection points of $f^*$ in $(-1, 1)$

$C(m, f^*) =$ number of distinct bias clusters at stationarity.

$k =$ number of sign-changing inflection points of the **target** $f^*$.

$f^* =$ the target function (e.g. $\sin(3\pi x)$).

**Sub-problems we address:**

- Can we prove $C(m, f^*) \leq k$ independently of $m$?
- Which neurons survive collapse to become cluster representatives?

# Experimental Design

### Nine target functions:

| $f^*(x)$ | $k$ | type |
|---|---|---|
| $\sin(\pi x)$ | 1 | trig |
| $x^3$ | 1 | poly |
| $\sin(2\pi x)$ | 3 | trig |
| $x^5 - 3x^3$ | 3 | poly |
| $\sin(3\pi x)$ | 5 | trig |
| $\sin(4\pi x)$ | 7 | trig |
| $\sin(5\pi x)$ | 9 | trig |
| $\sin(6\pi x)$ | 11 | trig |
| $\sin(7\pi x)$ | 13 | trig |

Trig: $k = 2n - 1$. Poly: irregular inflection spacing.

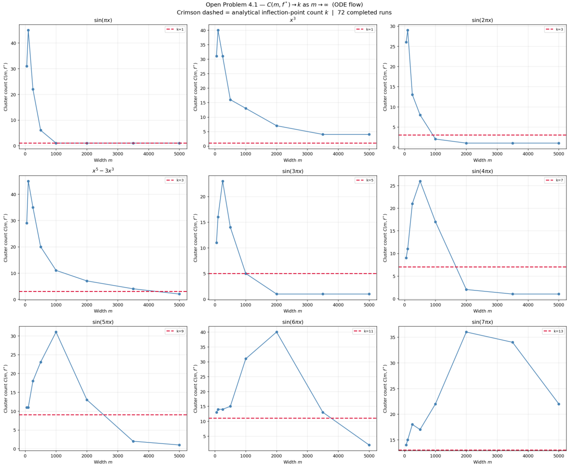
### Main sweep:

- $m \in \{50, 100, 250, 500, 1000, 2000, 3500, 5000\}$: number of neurons in the network
- $T = 500$: time horizon for ODE integration. SciPy's RK45 solver, a standard adaptive step-size integrator, numerically solves the gradient flow equations from $t = 0$ to $t = T$, automatically adjusting step size to maintain accuracy.
- **72 completed runs** across 9 targets

**Discrete GD comparison**

Fixed step-size discrete GD did not reproduce $C \to k$ for most targets. Adaptive step sizing (as in RK45) appears to matter for the clustering behavior to emerge.

# OP 4.1: $C(m, f^*)$ Across All 9 Targets




$C(m, f^*)$ vs. $m$ at $T = 500$ (72 runs, 9 targets). **Crimson dashed line** $= k$. All panels show rise-then-fall. Low-$k$ trig targets reach $C = k$; $\sin(7\pi x)$ $(k = 13)$ has not yet confirmed $C = k$ at tested widths (stationarity unconfirmed at $m = 5000$).

# OP 4.1: The Rise-Then-Fall Dynamic

## Universal pattern across all 9 targets:

1. **Rise:** Small $m$ cannot populate all $k$ inflection attractors simultaneously. $C$ grows above $k$.
2. **Peak:** $C$ reaches a maximum scaling with $k$ (up to $C = 40$ for $k = 11$).
3. **Fall:** Collapse proceeds and $C$ decreases toward $k$.

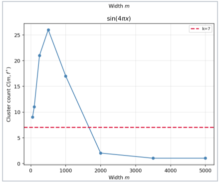
**Key observation**

The peak of $C(m)$ shifts to **larger** $m$ as $k$ grows. High-$k$ targets require much wider networks before the fall phase begins.

**Selected $C(m)$ at $T = 500$:**

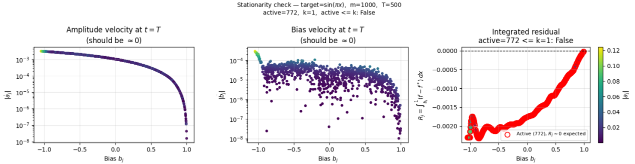
| Target | $k$ | $m$=1000 | $m$=2000 | $m$=5000 |
|---|---|---|---|---|
| $\sin(\pi x)$ | 1 | **1** | 1 | 1 |
| $\sin(3\pi x)$ | 5 | **5** | †1 | †1 |
| $\sin(4\pi x)$ | 7 | 17 | 2 | †1 |
| $\sin(7\pi x)$ | 13 | 22 | 36 | ‡**13** |
| $x^5 - 3x^3$ | 3 | 11 | 7 | 2 |
| $x^3$ | 1 | 13 | 7 | 4 |

† Cluster structure absent or too dense for fixed threshold to detect.
‡ $C = k$ but not fully stationary ($\max |\dot{a}_j| = 0.034$).




$\hookrightarrow \sin(4\pi x)$, $k = 7$: rise-then-fall

# OP 4.1: Stationarity and Fixed-Point Verification




$\sin(\pi x)$, $m = 1000$, $T = 500$

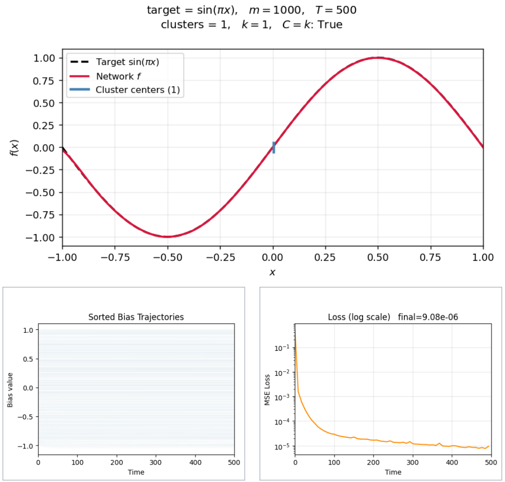
Final $(a_j, b_j)$ substituted back into the ODEs.

**Left:** $|\dot{a}_j| \approx 10^{-3}$–$10^{-5}$

**Center:** $|\dot{b}_j| \approx 10^{-4}$–$10^{-5}$

**Right:** $R_j \approx 0$ for active neurons (red), confirming the theoretical fixed-point condition numerically.

# OP 4.1 Confirmation: $\sin(\pi x)$, $k = 1$




$\sin(\pi x)$, $m = 1000$, $k = 1$

**Top:** Clean final fit vs. target. Blue ticks = single cluster center.

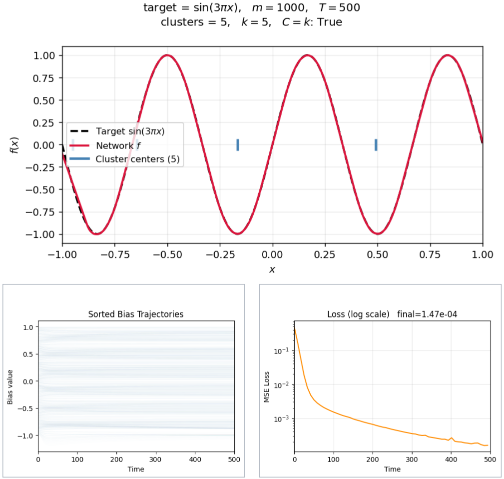
**Bottom:** Bias trajectories (all 1000 collapse to one location) and MSE loss on log scale.

**Strongest confirmation**

$C = 1 = k$ for *all* $m \geq 1000$ through $m = 5000$, fully stationary ($\max |\dot{a}_j| < 3.6 \times 10^{-3}$).

Only target with $C = k$ confirmed robustly across the full $m$ sweep.

# OP 4.1 Confirmation: $\sin(3\pi x)$, $k = 5$




$\sin(3\pi x)$, $m = 1000$, $k = 5$

**Top:** Clean final fit. Five blue ticks mark the 5 bias clusters, one per inflection point.

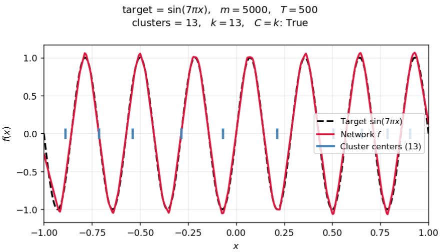
**Bottom:** Bias trajectories and MSE loss on log scale.

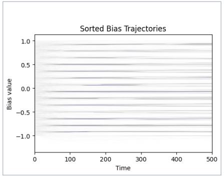
**Non-monotone path to $k$**

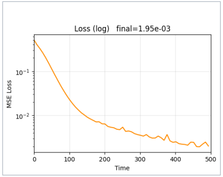
$$\underbrace{8}_{m=50} \to \underbrace{14}_{\text{peak},\ m=500} \to \underbrace{\mathbf{5}}_{k,\ m=1000} \to \underbrace{\dagger 1}_{m=2000}$$

Rises above $k$, reaches $k = 5$, then †cluster structure dissolves at larger $m$.

# OP 4.1: $\sin(7\pi x)$, $k = 13$ (Most Complex Target)




$\sin(7\pi x)$, $m = 5000$, $k = 13$

**Top/Bottom:** At $T = 500$, the fit shows 13 clusters aligning with inflection points, but the network has not settled.

### $C$ at $m = 5000$

$T = 500$: $C = 13 = k$, $\max|\dot{a}_j| = 0.034$

$T = 1000$: $C = 22$, $\max|\dot{a}_j| = 0.077$

$C = 13$ was **transient**: the run has not stabilized at $m = 5000$, $T = 500$.

### What is needed

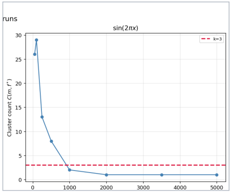
Confirmed $C \to k$ for $k = 13$ likely requires $m \gg 5000$. Larger $m$ provides the redundancy for stable collapse; longer $T$ alone is insufficient if $m$ is too small.

# OP 4.1: The Below-$k$ Phenomenon

**Three cases at verified stationary states with $C < k$. Re-run with 2 additional seeds each:**

| Target | $k$ | $m$ | seed 42 | seed 5 | seed 25 |
|---|---|---|---|---|---|
| $\sin(2\pi x)$ | 3 | 1000 | 2 | 1 | 1 |
| $\sin(4\pi x)$ | 7 | 2000 | 2 | 3 | 5 |
| $x^5 - 3x^3$ | 3 | 5000 | 2 | 2 | **3**$= k$ |

$C$ **varies across seeds** in all three cases. No single $C < k$ value is consistent.




$\hookrightarrow$ $\sin(2\pi x)$, $k = 3$: $C$ stays below $k$

## Interpretation

$C$ varies across seeds $\Rightarrow$ $C < k$ states are **initialization-dependent local minima**, not universal fixed points.

Gradient flow has multiple basins of attraction. The $k$-cluster basin exists but is not the only one.

## $x^5 - 3x^3$ seed 25: $C = k = 3$

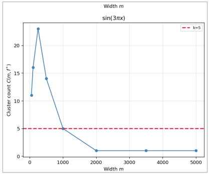
First time $x^5 - 3x^3$ reaches $C = k$. Shows the $k$-cluster attractor exists for polynomial targets, though harder to find.

# Instability Near $k$: Results

**Question:** Are states with $C \neq k$ unstable? Does gradient flow push them toward $C = k$?

**22 qualifying runs:** 8 ($C = k$), 8 ($C > k$), 6 ($C < k$). $T_{\text{perturb}} = 1000$.

| Category | Status | Finding |
|---|---|---|
| $C > k$ states | 8/8 done | **No change.** $C$ stays fixed; ODE fixed points do not dissolve. |
| $C < k$ states | 12/12 done | **No change.** Injected neuron amplitude stays at $0.01$ and cannot grow. |
| $C = k$ states | 16/16 done | **All returned to $k$.** Exact-$k$ states are stable attractors. |




$\hookrightarrow$ $\sin(3\pi x)$, $k = 5$: $C = k$ state is stable

**$C > k$: stable fixed points**

Continuing ODE for $T = 1000$: $C$ does **not** decrease. $C > k$ states are *stable* fixed points, not metastable.

**$C < k$: injection-resistant**

Injected neurons gain **zero amplitude**. ODE provides no restoring force toward $k$. Supports: $C < k$ states are true fixed points.

**Takeaway**

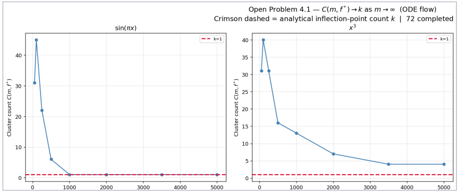
Both $C > k$ and $C < k$ states are stable fixed points. A proof of $C \to k$ cannot argue that other states dissolve on their own; it must explain why random initializations land at $C = k$.

# OP 4.1: Polynomial vs. Trigonometric Targets

## Same $k$, very different convergence speed:

| Target | $k$ | $m$=1000 | $m$=2000 | $m$=5000 |
|---|---|---|---|---|
| $\sin(\pi x)$ | 1 | **1** | 1 | 1 |
| $x^3$ | 1 | 13 | 7 | 4 |
| $\sin(2\pi x)$ | 3 | 2 | †1 | †1 |
| $x^5-3x^3$ | 3 | 11 | 7 | 2 |

$x^3$ reaches only $C = 4$ at $m = 5000$, despite $k = 1$.




$\hookrightarrow$ $\sin(\pi x)$ (left) vs $x^3$ (right), same $k = 1$

### Why inflection geometry matters

For $\sin(n\pi x)$, the $k$ inflection points are *evenly spaced* across $[-1, 1]$, forming strong, symmetric attractors that neurons organize around early in training.

Polynomial targets have *irregular* spacing: weaker, asymmetric attractors. Neurons take far longer to find them.

**Implication:** a proof of OP 4.1 must account for attractor geometry, not just count.

# OP 4.1: Key Findings

**What was confirmed**

- $C \to k$ confirmed for $\sin(\pi x)$ ($k = 1$), all $m \geq 1000$ (strongest case).
- $\sin(7\pi x)$: $C = 13$ at $m = 5000$, $T = 500$ was transient; $C = 22$ at $T = 1000$. Confirmed $C \to k$ requires $m \gg 5000$.
- $C = k$ states are stable attractors: all 16 perturbation runs returned to $C = k$.

**What the data reveals about OP 4.1**

- $C < k$ local minima exist for 3 targets at genuinely stationary states.
- $C > k$ and $C < k$ states are both stable fixed points; a proof of $C \to k$ must rely on **initialization geometry**, not instability.
- Polynomial targets converge far more slowly due to irregular attractor geometry.

# Open Problem 4.2: Higher-Dimensional Collapse

### Question

For a structured target $f^* : [-1, 1]^2 \to \mathbb{R}$, does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

**2D network:**

$$f(x) = \sum_{j=1}^{m} a_j \, \sigma(w_j^\top x + b_j), \quad x \in [-1, 1]^2$$

Each neuron defines an oriented hyperplane (line in 2D). Scale-invariant: $\theta_j = \angle(w_j)$, $\beta_j = b_j / \|w_j\|$.

**Collapse in 2D:** the $(\theta_j, \beta_j)$ cloud concentrates onto a low-dimensional subset.

### Diagnostic

Angular entropy $H(\{\theta_j\})$: near 0 if collapsed; near $\log(36) \approx 3.58$ if uniform. Effective neurons $N_{\text{eff}} = e^H$:
$N_{\text{eff}} = 1$ full collapse; $N_{\text{eff}} \approx 36$ uniform.

### Width sweep

$m \in \{64, 256, 512, 1024\}$, 6000 epochs, Adam.

### Three targets studied

Ridge, Separable, and Radial.
See next slide for expected vs. observed behavior.

# Open Problem 4.2: Target Functions

**Ridge**

$\sin(2\pi u^\top x) + 0.5\sin(4\pi u^\top x)$

*Expected (small width):*
concentration near $\theta \approx 53^\circ$

*Observed:*
concentration weakens as width increases

**Separable**

$\sin(2\pi x_1) + 0.3\sin(2\pi x_2)$

*Expected (small width):*
two dominant direction families

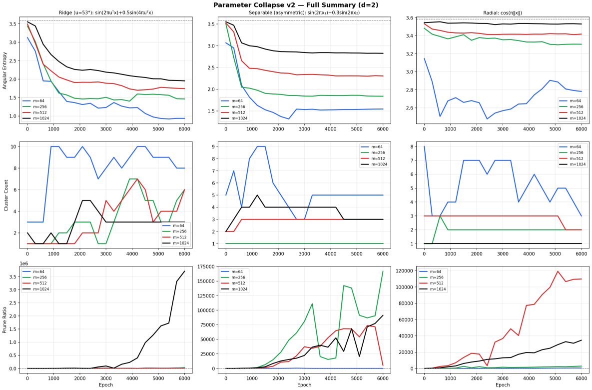
*Observed:*
directional structure becomes diffuse at large width

**Radial (reference)**

$\cos(\pi\|x\|)$

*Reference case:*
approximately isotropic orientation distribution

# OP 4.2: Orientation Distributions




### Entropy $H$ / Eff. neurons $N_{\text{eff}}=e^H$ at epoch 6000:

| Target | $m=64$ | $m=256$ | $m=512$ | $m=1024$ |
|---|---|---|---|---|
| Ridge | 0.94 / 3 | 1.46 / 4 | 1.74 / 6 | 1.95 / 7 |
| Separable | 1.55 / 5 | 1.84 / 6 | 2.31 / 10 | 2.83 / 17 |
| Radial | 2.78 / 16 | 3.31 / 27 | 3.42 / 31 | **3.53 / 34** |

$N_{\text{eff}}=1$: full collapse; $N_{\text{eff}}\approx 36$: uniform. Max $H=\log(36)\approx 3.58$

**Open question**

Why does directional concentration weaken as width grows? Observed collapse appears strongest in moderately overparameterized networks. The mechanism behind this trend remains unclear.

# OP 4.2: Key Findings

**What the data shows**

- At small width, directional concentration reflects target geometry:
  ridge $\to$ one direction family;
  separable $\to$ two families;
  radial $\to$ broadly distributed.
- Concentration decreases steadily as width grows.
- Networks increasingly fit targets using many distinct directions.

**Effective directional complexity**

$N_{\text{eff}} = e^H$ measures how many distinct directions the network actively uses.

- $N_{\text{eff}}$ increases monotonically with width.
- Wider networks rely less on directional collapse.
- Radial $m{=}1024$: $N_{\text{eff}} \approx 34$ of 36 (nearly uniform).

# Open Problem 4.3: Pruning Bound

## Bound (from the slides)

$$\left\|\tilde{f} - f\right\|_{L^2} \leq \delta \cdot \sum_{j=1}^{m} |a_j|$$

**Setup:** After gradient flow, bias clusters have formed. *Prune* by replacing each cluster with a single representative neuron, giving $\tilde{f}$ with only $k$ neurons.

$\delta =$ maximum intra-cluster diameter (how spread out the biases are within each cluster).

$\sum |a_j| =$ total weight mass of the full $m$-neuron network.

**Why it matters:** Provides a formal $L^2$ certificate that collapsing from $m$ neurons to $k$ neurons is safe, bridging the collapse phenomenon and approximation theory.

## Why the bound holds

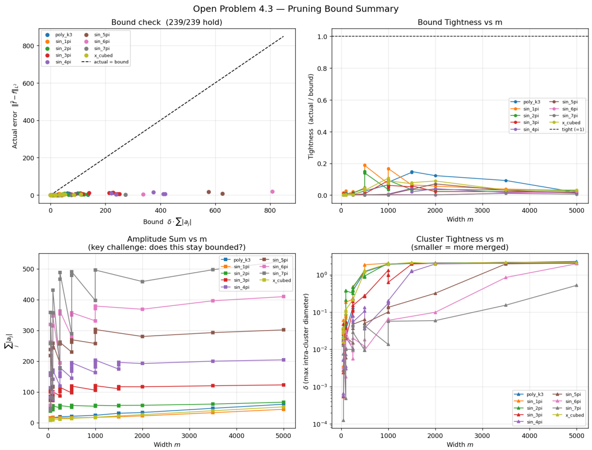
Within a cluster, all $b_j$ differ by at most $\delta$, so $\sigma(x - b_j) \approx \sigma(x - b_{\text{center}})$ up to an error $\leq \delta$. Summing over $m$ neurons, errors accumulate at most as $\delta \cdot \sum |a_j|$.

## The challenge

For the bound to be *non-vacuous* (goes to 0 as networks grow), we need $\delta \cdot \sum |a_j| \to 0$.

Our data shows $\sum |a_j| \sim \sqrt{m}$. So we need $\delta \to 0$ fast enough, which must come from longer training $T$, not wider networks $m$.

# OP 4.3: Pruning Bound Verification




**Bound:** $\left\|\tilde{f} - f\right\|_{L^2} \leq \delta \cdot \sum_j |a_j|$

**72 / 72 runs: bound holds**

Tightness (actual / bound): $0.0002$ to $0.148$. Substantial slack in all cases.

**Key obstacle:**

- $\sum_j |a_j| \sim \sqrt{m}$ empirically
- $\delta$ does not shrink with $m$ at $T = 500$
- Bound loosens $\sim \sqrt{m}$ while actual error plateaus

**Path to non-vacuousness**

$\delta \to 0$ requires $T \to \infty$, *not* $m \to \infty$. Study $\delta(T)$ at fixed $m$ to make the bound formally useful.

# Lottery Ticket: Which Neurons Survive?

**Setup:** At $t = 0$, amplitudes $a_j \sim \mathcal{N}(0, 0.01)$ are random. Only the bias placement $b_j$ is structured. Neurons near an inflection point are *geometrically lucky*.

**Five conditions per (target, $m$):** `full`, `geometric`, `bias_only`, `amp_only`, `random_k`

52 runs; 4 targets, $m \in \{500, 1000, 1500\}$.

**Claim 2 (performance): FALSE**

Lucky $k$ neurons alone achieve $100\times$–$50{,}000\times$ higher loss than the full network. $k$ kinks cannot approximate a smooth target; full redundancy is necessary.

**Claim 1 (survival): bias position is key**

**geometric** $\approx$ **bias_only** in every case (loss diff $< 0.002$). Initial amplitudes carry *no information*.

**amp_only** is the worst condition: good amplitudes with random biases performs worse than good biases with random amplitudes.

**Conclusion:**
Bias position at $t = 0$ is the sole informative quantity for collapse. The "winning ticket" is a good bias placement, not a good amplitude.

# Open Questions and Next Steps

**OP 4.1 (open):**

- A proof of $C \to k$ must account for initialization geometry. Below-$k$ local minima exist, so the $k$-cluster basin must be shown to dominate at large $m$
- **4.1.1:** $C > k$ states are *stable* fixed points, so instability alone cannot drive $C \to k$

**OP 4.2 (open):**

- Why does directional concentration weaken as width grows? The mechanism behind this trend remains unclear.

**OP 4.3, path to a formal proof:**

- Study $\delta(T)$ at fixed $m$: does $\delta \to 0$ as $T \to \infty$?
- Exponent $\alpha$ in $\sum_j |a_j| = O(m^\alpha)$; data suggests $\alpha \approx 1/2$

**Immediate priorities**

(1) $\delta(T)$ trajectory study
(2) Theory for capacity-constrained collapse

# Conclusion

**What we showed:**

- $C \to k$ confirmed for $\sin(\pi x)$ ($k = 1$), $m \geq 1000$.
- **Rise-then-fall** is universal: peak scales with $k$; inflection geometry sets the convergence rate.
- **Instability:** $C > k$ and $C < k$ states are stable fixed points. Proof of $C \to k$ must use initialization geometry.
- **OP 4.2:** Effective directional complexity grows with width; wider networks rely less on orientation collapse than narrower ones.
- **OP 4.3:** 72 / 72 hold. $\sum |a_j| \sim \sqrt{m}$. Non-vacuousness requires $\delta \to 0$ as $T \to \infty$.
- **Lottery ticket:** Bias position at $t = 0$ is sole informative quantity. Amplitude values irrelevant. Performance claim false.

**Broader implication**

Gradient flow is not only an optimizer; it is a **compression algorithm** that discovers the minimum-complexity representation of the target's intrinsic structure.

Inflection points of $f^*$ act as **fixed-point attractors**: neurons near one survive; the rest dissolve.

A rigorous proof would explain generalization in overparameterized networks.

## Thank you

**Repository & documentation** `github.com/calebabr/NSF-RTG-Project4` full scripts, data, and result tables in `README.md`